# LESSON SIX

# HOSTING A WEBSITE ON GITHUB AND USING A CUSTOM DOMAIN NAME

## 6.1 Meaning of Website Hosting

I learned that website hosting is the process of making a website available on the internet so that people can access it through a web browser.

When I develop a website on my computer, other people cannot normally access it through the internet. To make it publicly accessible, I need to upload it to a hosting platform.

## 6.2 Meaning of GitHub Pages

I learned that GitHub Pages is a free hosting service provided by GitHub. It allows developers to publish websites directly from a GitHub repository.

It is mainly used for static websites developed with:

- HTML
- CSS
- JavaScript

For example, I can develop a personal portfolio website using HTML, CSS and JavaScript, upload it to GitHub and publish it using GitHub Pages.

## 6.3 Steps for Hosting a Website on GitHub Pages

I learned that I can host a static website on GitHub Pages by following these steps:

**Step 1: Create a GitHub Account**

I first need to create an account on GitHub if I do not already have one.

**Step 2: Create a New Repository**

I create a new repository and give it a suitable name.

**Step 3: Upload Website Files**

I upload my website files, such as:

- index.html
- style.css
- script.js
- Images and other resources

The main homepage should normally be named index.html.

**Step 4: Open Repository Settings**

I open the repository and select Settings.

**Step 5: Select Pages**

Under the repository settings, I select Pages.

**Step 6: Select the Publishing Source**

I choose the branch containing my website files, usually main, and select the root folder if the index.html file is located there.

**Step 7: Save and Publish**

I save the settings and wait for GitHub Pages to publish the website.

GitHub provides a website address that I can share with other people.

## 6.4 Meaning of a Custom Domain Name

I learned that a custom domain name is a unique website address chosen by the owner of a website.

For example:

- `myportfolio.com`
- `mybusiness.com`
- `www.mycompany.com`

A custom domain makes a website look more professional and easier to remember than a default hosting address.

## 6.5 How to Connect a Custom Domain to GitHub Pages

I learned that I can connect a custom domain to my GitHub Pages website by following these steps:

1. Purchase a domain name from a domain registrar.
2. Open my GitHub repository.
3. Go to Settings.
4. Select Pages.
5. Enter my custom domain name in the Custom domain section.
6. Save the changes.
7. Open the DNS settings of my domain provider.
8. Add the DNS records required by GitHub Pages.
9. Wait for the DNS changes to take effect.
10. Enable HTTPS in the GitHub Pages settings when it becomes available.

## 6.6 Meaning of DNS

I learned that DNS means Domain Name System. It is a system that connects domain names to the servers or services responsible for hosting websites.

For example, when someone enters `www.mycompany.com` into a browser, DNS helps direct the browser to the correct website.

For GitHub Pages, the DNS records must be configured correctly with the domain provider.

## 6.7 Important Limitation of GitHub Pages

I learned that GitHub Pages is designed for static websites. It supports HTML, CSS and JavaScript, but it does not directly execute server-side programming languages such as PHP or connect to a MySQL database.

Therefore, if I develop a dynamic application using PHP and MySQL, I need a different hosting service that supports PHP and databases.

Examples include shared web hosting services that provide PHP and MySQL support.

## 6.8 Advantages of GitHub Pages

I learned that GitHub Pages has several advantages:

1. It provides free hosting for supported static websites.
2. It connects directly to GitHub repositories.
3. It makes it easy to publish project portfolios.
4. It supports custom domain names.
5. It supports HTTPS.
6. It allows website updates through GitHub.

## Conclusion

From this lesson, I understood how to publish a static website using GitHub Pages and how to connect a custom domain name. I also learned the difference between static website hosting and dynamic hosting, especially when developing applications that require PHP and MySQL.